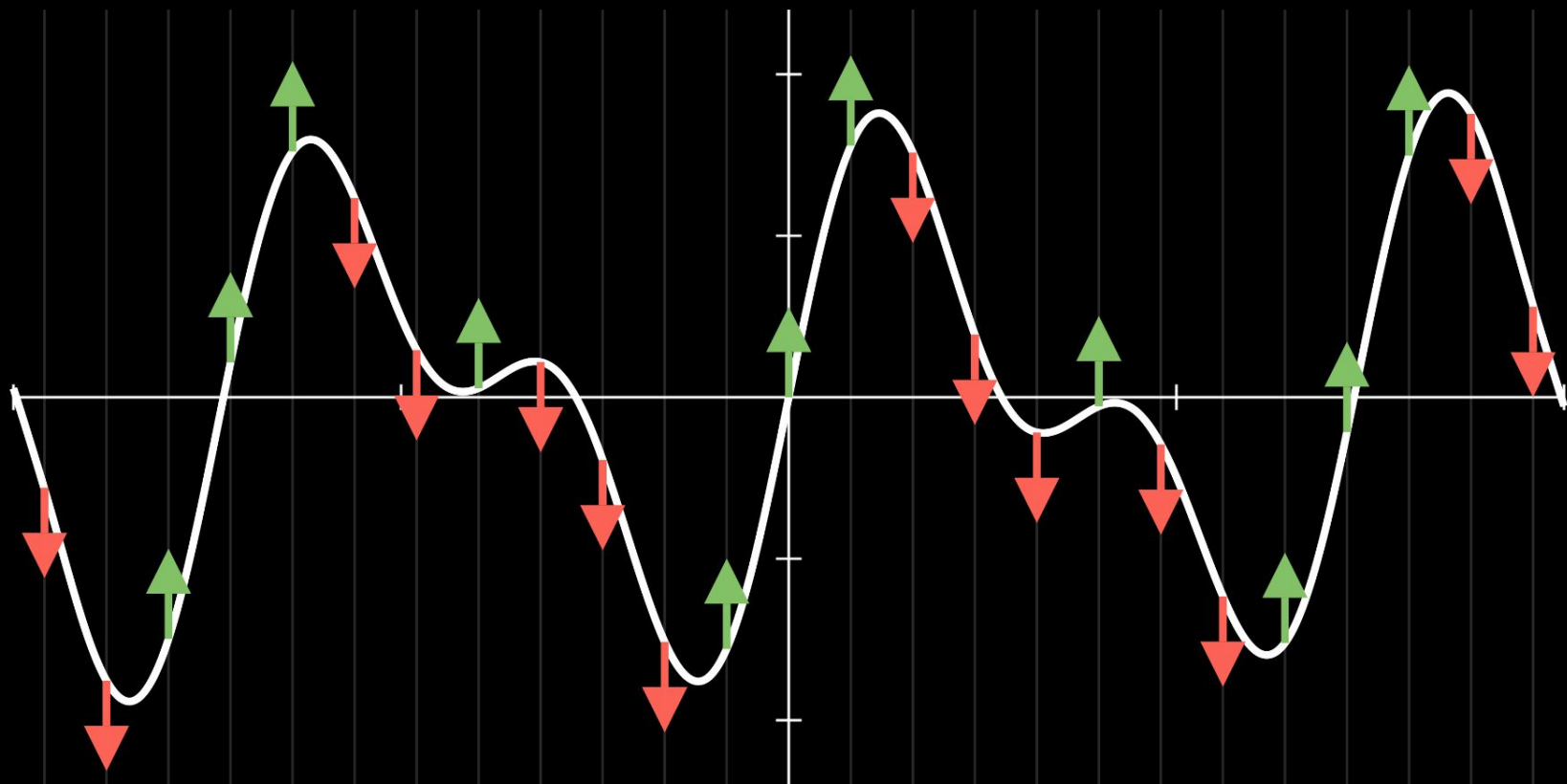
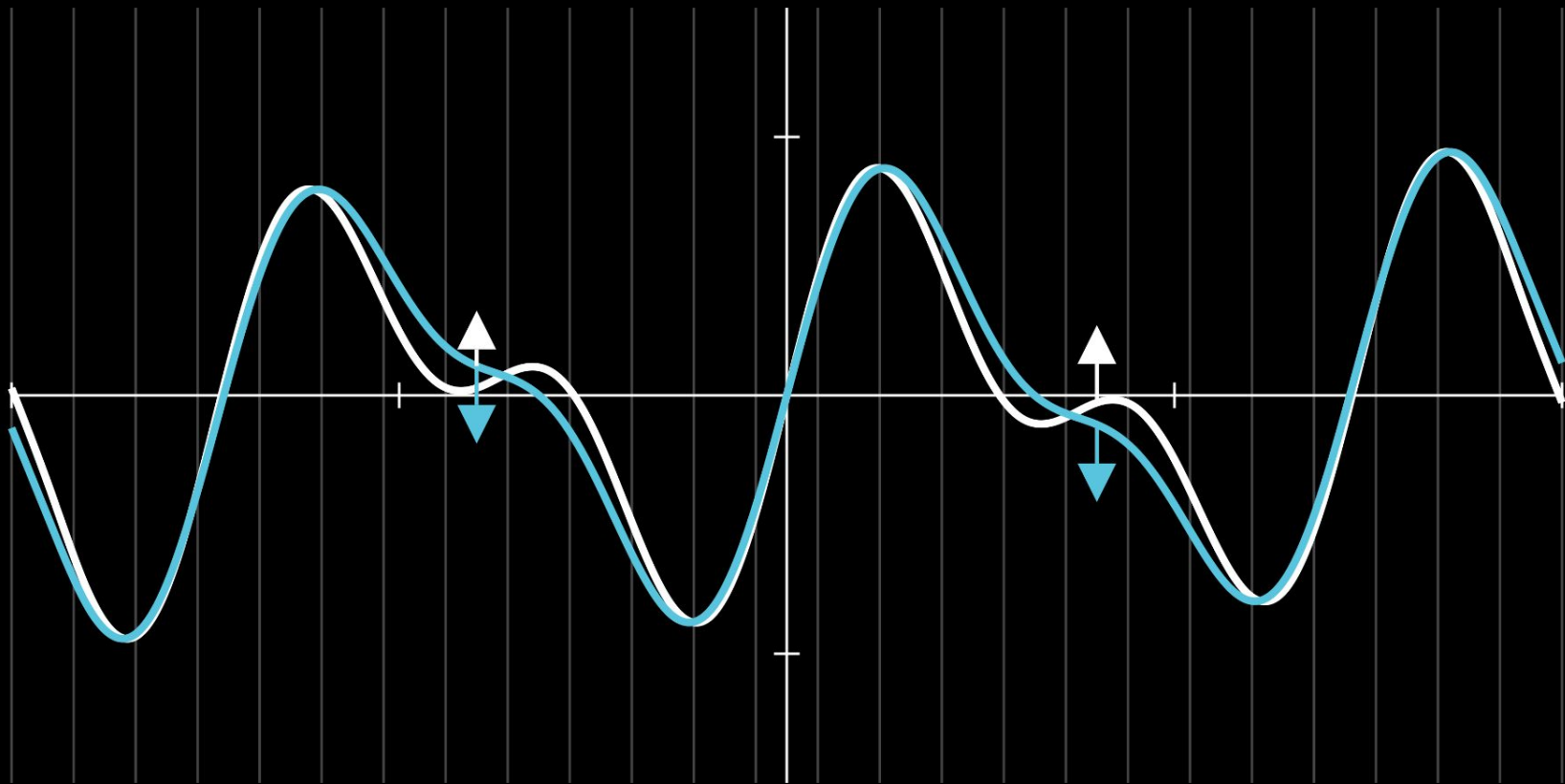


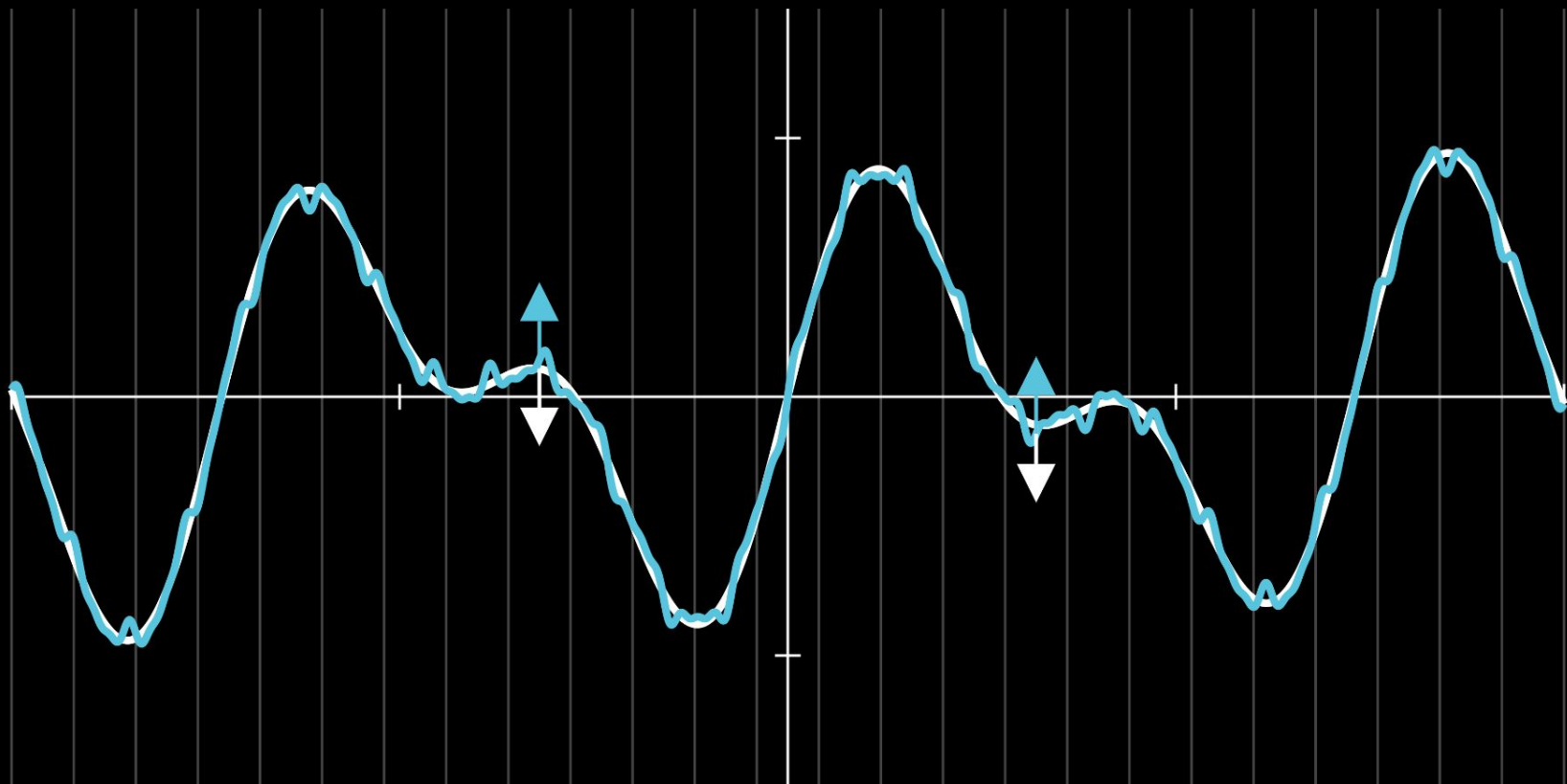
Directional Sign Loss

A Topology-Preserving Loss Function that Approximates the Sign of Finite Differences

Harvey Dam, Tripti Agarwal, Ganesh Gopalakrishnan

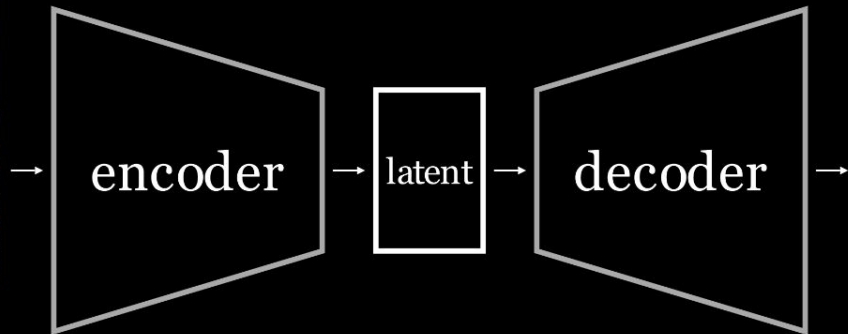






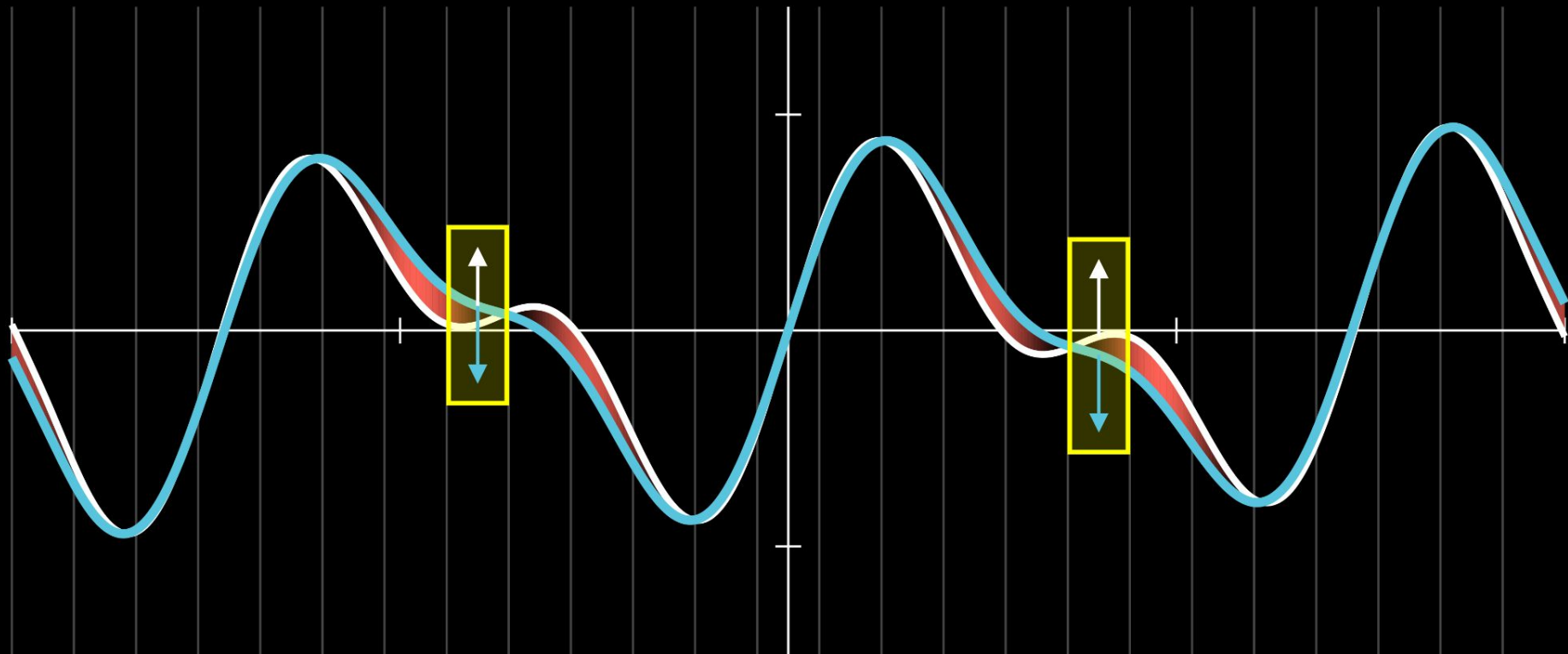


input

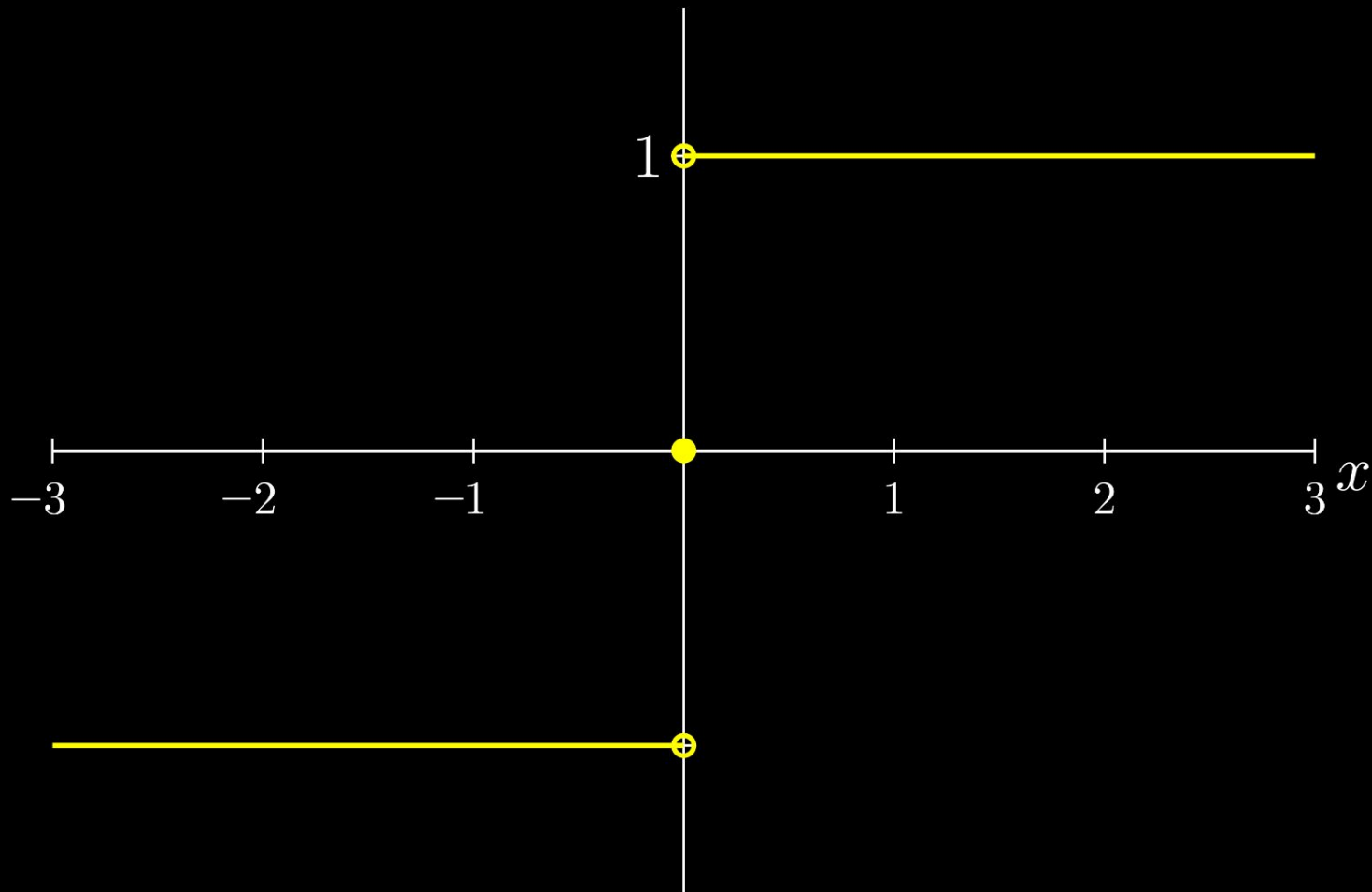


output

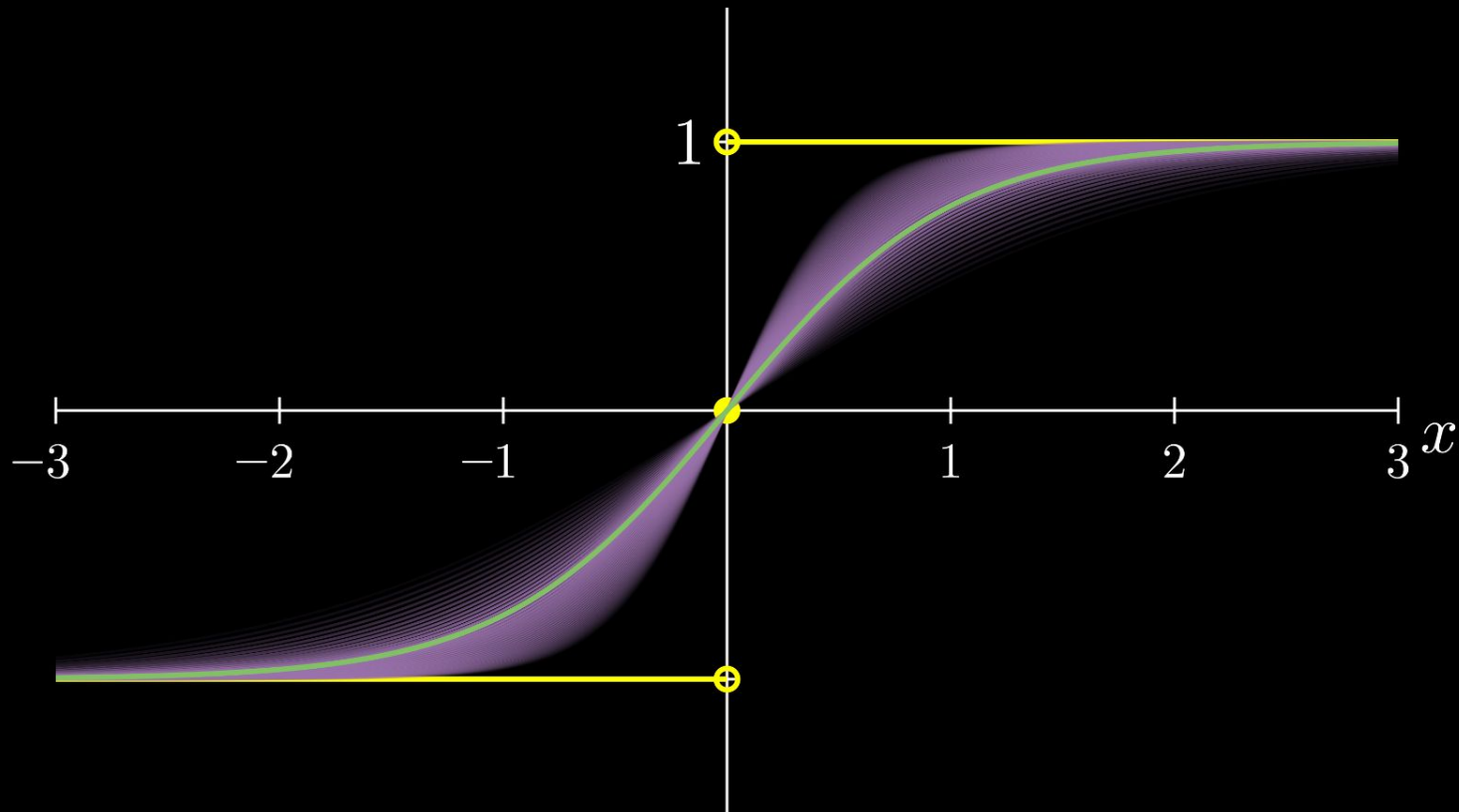
$$\text{loss} = \text{MSE} + \sum \mathbb{I}(\text{sign}(D_{\text{original}}) \neq \text{sign}(D_{\text{reconstruction}}))?$$



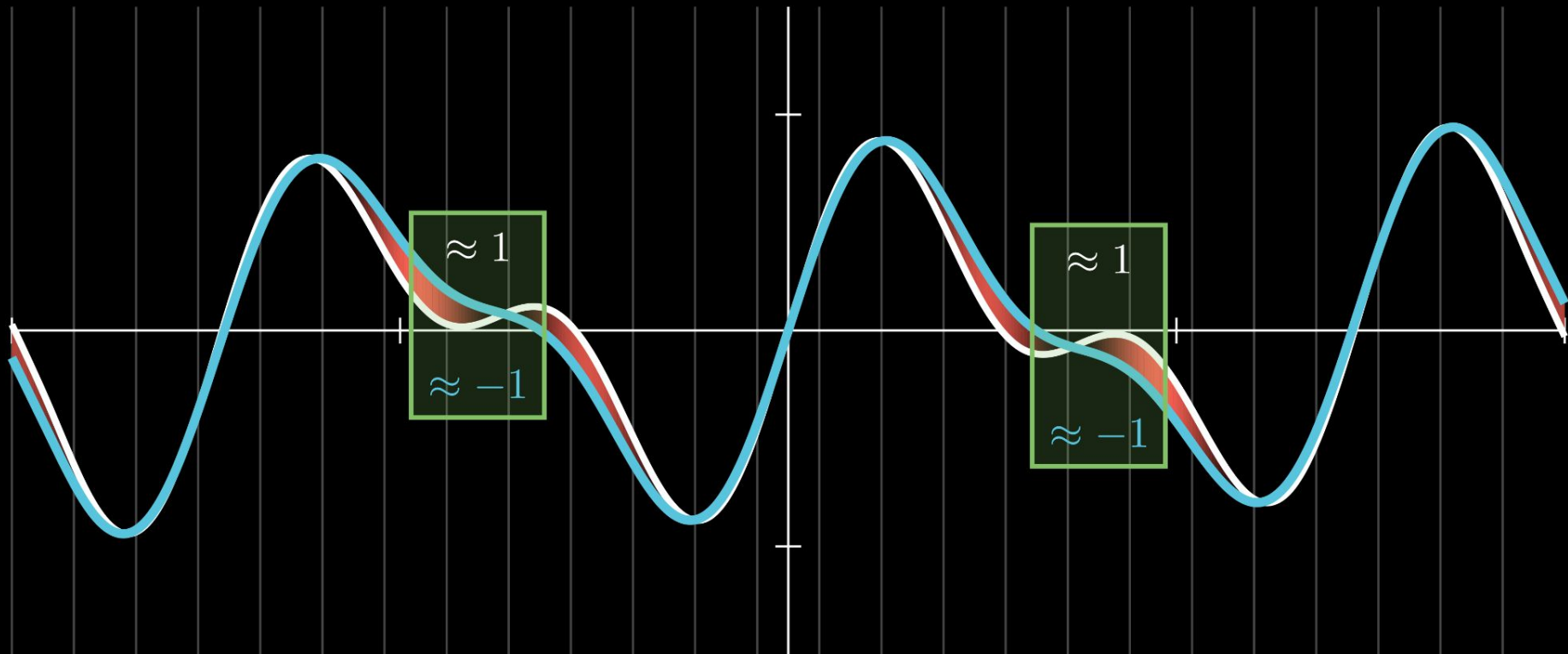
$\text{sign}(x)$



$\text{sign}(x)$ vs $\tanh(sx)$



$$\text{loss} = \text{MSE} + \sum \mathbb{I}(\text{sign}(D_{\text{original}}) \neq \text{sign}(D_{\text{reconstruction}})) \\ \left| \tanh(sD_{\text{original}}) - \tanh(sD_{\text{reconstruction}}) \right|$$



directional sign loss (DSL) pseudocode

Input: $X, \hat{X}$ shaped $(n_1, n_2, \dots, n_d)$, sharpness s , optional weights W

Output: ℓ

$\ell \leftarrow 0$

for $i \leftarrow 1$ to d :

 Compute finite differences:

$$D_X \leftarrow X_{(:,*)*(i-1),2:n_i,(:,*)*(d-i)} - X_{(:,*)*(i-1),1:(n_i-1),(:,*)*(d-i)}$$

$$D_{\hat{X}} \leftarrow \hat{X}_{(:,*)*(i-1),2:n_i,(:,*)*(d-i)} - \hat{X}_{(:,*)*(i-1),1:(n_i-1),(:,*)*(d-i)}$$

 Find approximate sign mismatches:

$$\Delta \leftarrow |\tanh(sD_X) - \tanh(sD_{\hat{X}})| / 2$$

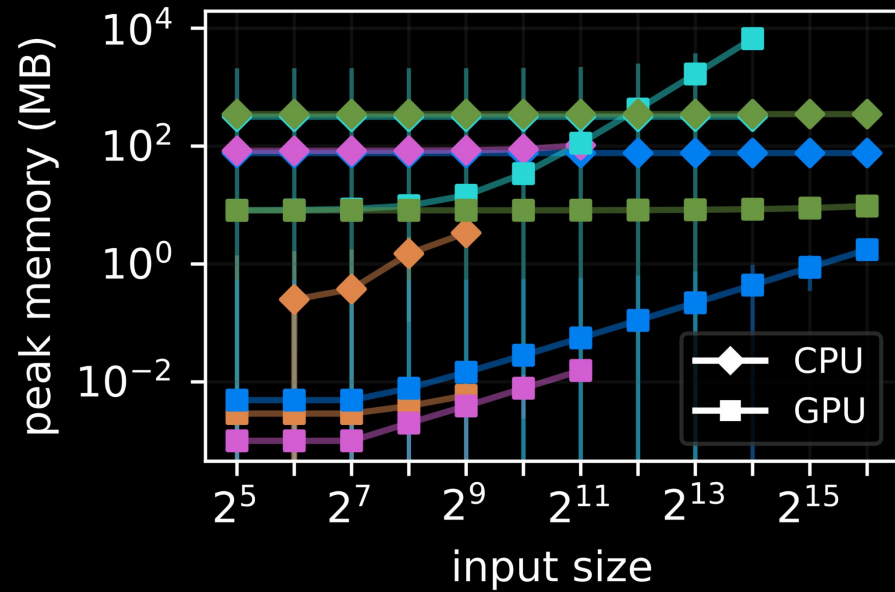
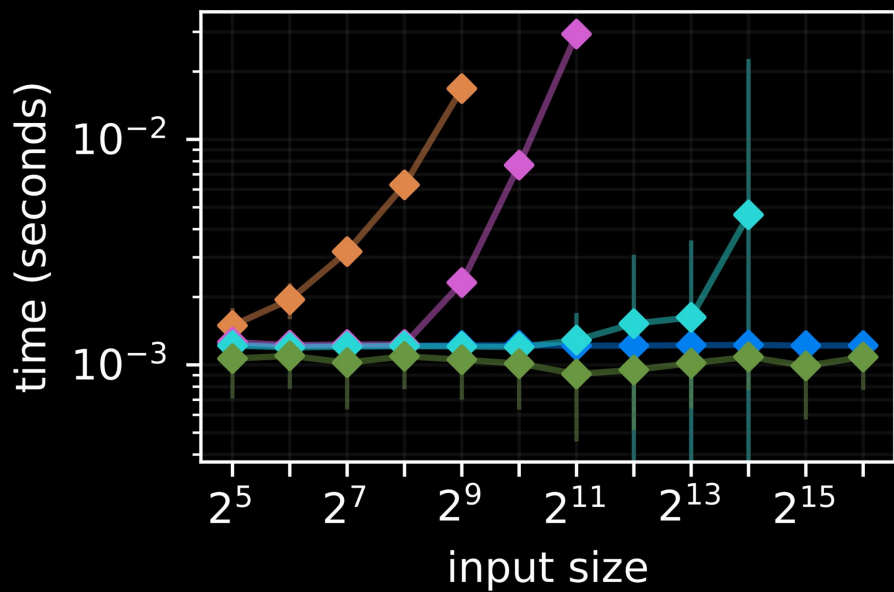
 With optional weights W , accumulate mismatches:

$$\ell \leftarrow \ell + (W_i \text{ or } 1) \sum \Delta$$

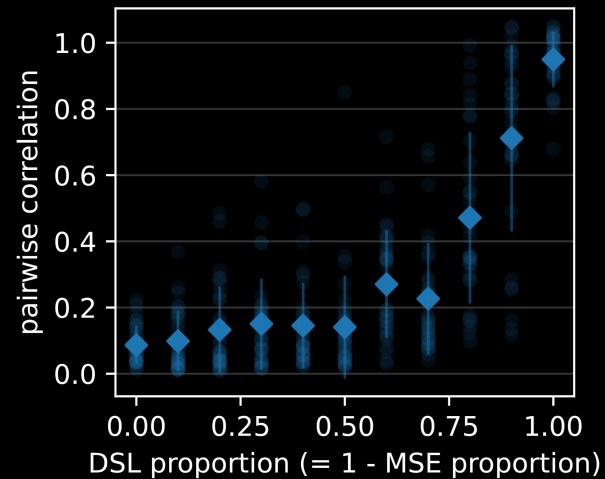
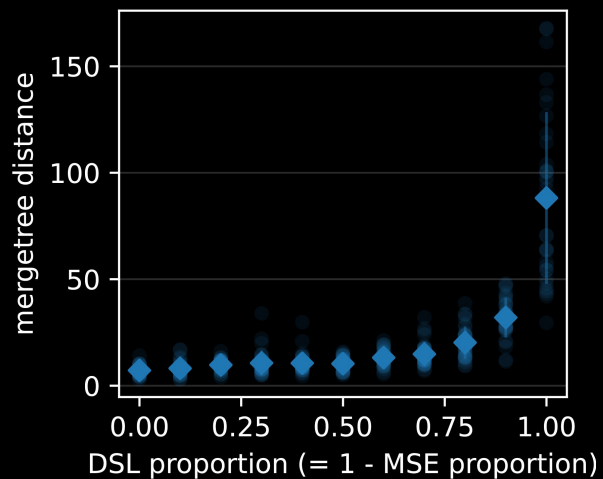
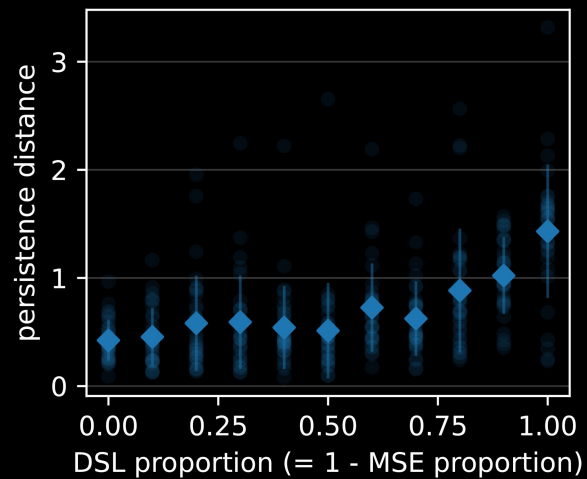
return ℓ

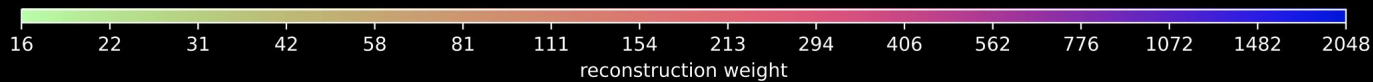
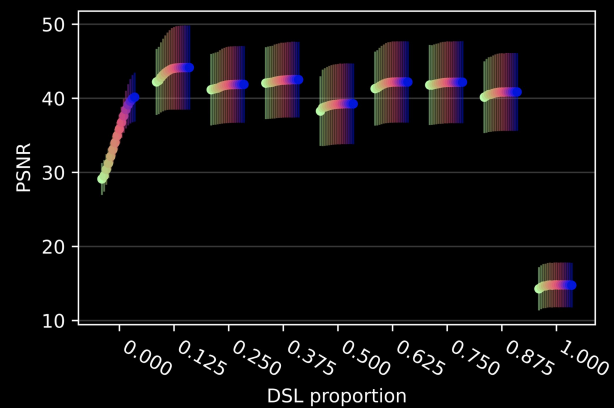
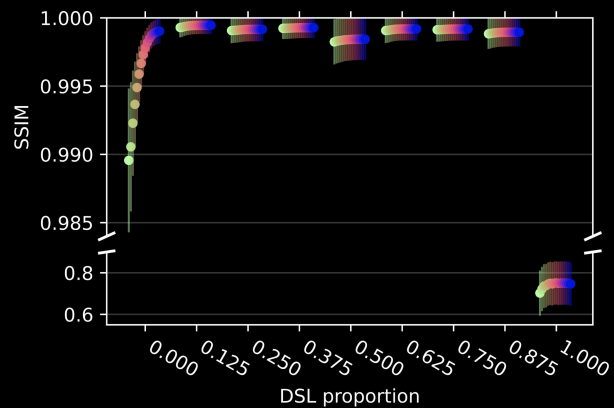
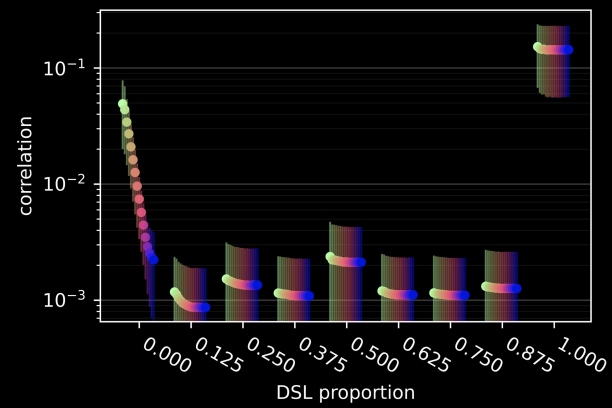
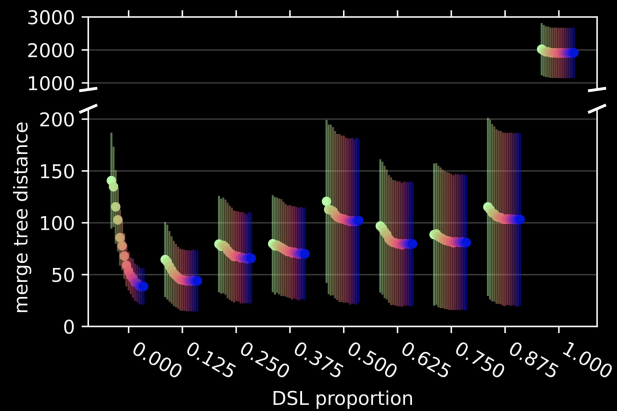
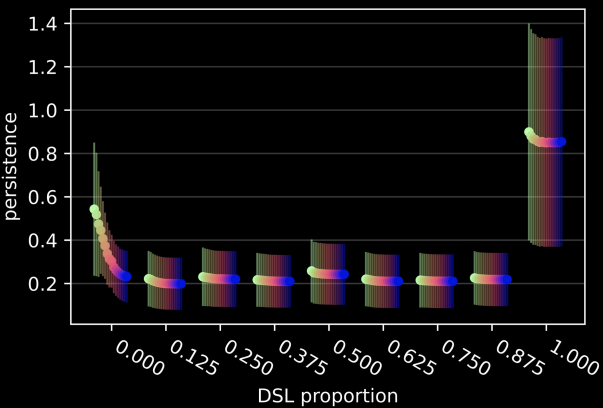
time and memory complexity

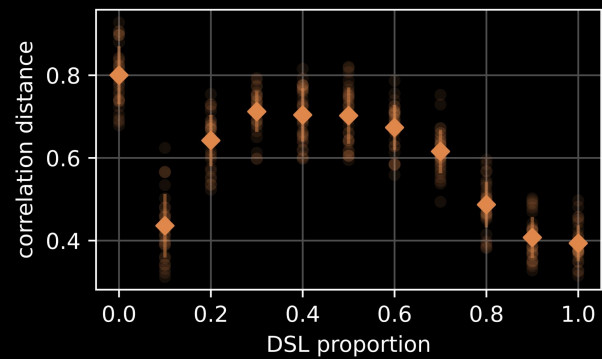
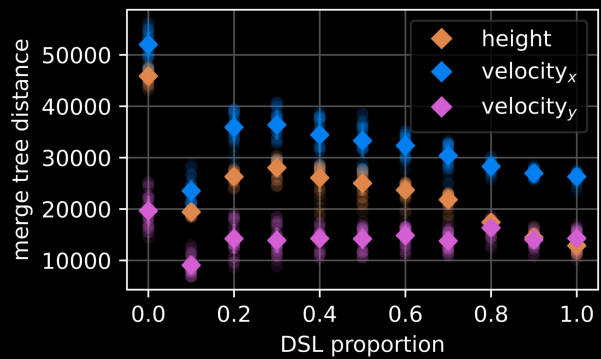
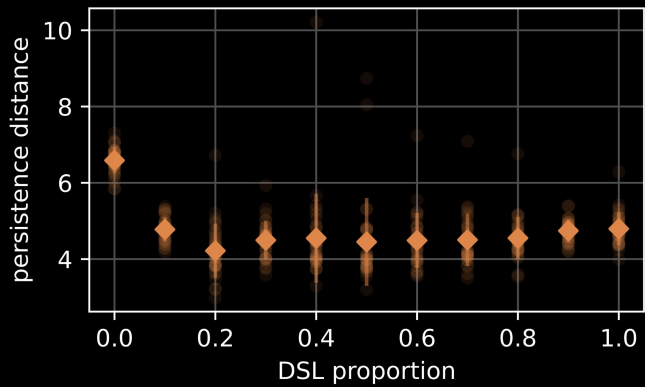
	less time	less memory
time complexity	$O\left(\log_2 d + \sum_{i=1}^d \log_2 n_i\right)$	$O\left(\textcolor{red}{d} \sum_{i=1}^d \log_2 n_i\right)$
memory complexity	$O\left(\textcolor{red}{d} \prod_{i=1}^d n_i\right)$	$O\left(\prod_{i=1}^d n_i\right)$



merge tree distance DSL persistence distance correlation distance MSE







Summary

Directional sign loss (DSL) is a loss function between same-shaped arrays that approximately penalizes mismatches between the signs of their finite differences.

It behaves like other measures of topological dissimilarity in terms of critical points while being parallelizable and differentiable.

It can be used for training autoencoders to preserve topological features.

Preprint: <https://arxiv.org/abs/2504.04202>